Derivation of the gradient (classification)

$$e^{(\alpha)} = -\sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln y_k(\underline{x}^{(\alpha)}; \underline{w})$$

$$y_k(\underline{x}^{(\alpha)}; \underline{w}) = \frac{\exp h_k(\underline{x}; \underline{w})}{\sum_l \exp h_l(\underline{x}; \underline{w})}$$

$$\frac{\partial e^{(\alpha)}}{\partial \underline{w}} = -\sum_{k=1}^{c} \frac{y_{Tk}^{(\alpha)}}{y_k(\underline{x}^{(\alpha)}; \underline{w})} \cdot \frac{\partial y_k(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}}$$

$$= -\sum_{k=1}^{c} \frac{y_{Tk}^{(\alpha)}}{y_k(\underline{x}^{(\alpha)}; \underline{w})} \left\{ y_k(\underline{x}^{(\alpha)}; \underline{w}) \frac{\partial h_k(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}} - \frac{\exp h_k(\underline{x}^{(\alpha)}; \underline{w})}{\left(\sum_l \exp h_l(\underline{x}^{(\alpha)}; \underline{w})\right)^2} \sum_{l=1}^{c} \exp h_l(\underline{x}^{(\alpha)}; \underline{w}) \cdot \frac{\partial h_l(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}} \right\}$$

$$= -\sum_{k=1}^{c} \frac{y_{Tk}^{(\alpha)}}{y_k(\underline{x}^{(\alpha)}; \underline{w})} \left\{ y_k(\underline{x}^{(\alpha)}; \underline{w}) \cdot \frac{\partial h_k(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}} - \underbrace{\frac{\exp h_k(\underline{x}^{(\alpha)}; \underline{w})}{\sum_l \exp h_l(\underline{x}^{(\alpha)}; \underline{w})}}_{y_k(\underline{x}^{(\alpha)}; \underline{w})} \cdot \sum_{l=1}^{c} \underbrace{\frac{\exp h_l(\underline{x}^{(\alpha)}; \underline{w})}{\sum_m \exp h_m(\underline{x}^{(\alpha)}; \underline{w})}}_{y_l(\underline{x}^{(\alpha)}; \underline{w})} \cdot \frac{\partial h_l(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}} \right\}$$

$$= -\sum_{k=1}^{c} y_{Tk}^{(\alpha)} \cdot \frac{\partial h_k(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}} + \underbrace{\left(\sum_{k=1}^{c} y_{Tk}^{(\alpha)}\right)}_{=1} \sum_{k=1}^{c} y_k(\underline{x}^{(\alpha)}; \underline{w}) \cdot \frac{\partial h_k(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}}$$

$$\frac{\partial e^{(\alpha)}}{\partial \underline{w}} = \sum_{k=1}^{c} \underbrace{\left(y_k(\underline{x}^{(\alpha)}; \underline{w}) - y_{Tk}^{(\alpha)} \right)} \cdot \underbrace{\frac{\partial h_k(\underline{x}^{(\alpha)}; \underline{w})}{\partial \underline{w}}}_{\rightarrow \text{ via backpropagation (for MLP)}}$$